
Lecture Notes

on Conditional Probability

by Arnaja Mitra

1 Conditional Probability

Let A and B be two events with $P(B) > 0$. The **conditional probability** of event A under the condition that event B has occurred is given by:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{--- (1)}$$

$P(B) > 0,$
 $0 \leq P(B) \leq 1$

Similarly,

$$P(B|A) = \frac{P(B \cap A)}{P(A)} \quad \text{--- (2)}$$

$P(A) > 0.$

Note:

$$P(A \cap B) = P(B \cap A)$$

$$P(A|A) = \frac{P(A \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$$

$$P(\bar{A}|B) = 1 - P(A|B), \quad \text{but} \quad P(A|\bar{B}) \neq 1 - P(A|B)$$

From (1) and (2), one can have:

$$P(A \cap B) = P(A|B) P(B), \quad \text{provided } P(B) > 0,$$

and

$$P(A \cap B) = P(B|A) P(A), \quad \text{provided } P(A) > 0.$$

$P(B) P(A|B) = \frac{P(A \cap B)}{P(B)} P(B)$
 $\cancel{P(B)}$

$P(A) P(B|A) = \frac{P(A \cap B)}{P(A)} P(A)$
 $\cancel{P(A)}$

2 Independent Events

Definition: Let A and B be two events in a sample space S with $P(A) > 0$, $P(B) > 0$. If the occurrence of event B does not change the probability of occurrence of event A , and vice versa, then the events A and B are called **independent events**.

Properties: If A and B are independent events, then

$$P(A \cap B) = P(A) P(B).$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)}$$

Also, if A and B are independent:

$$P(A|B) = P(A),$$

and

$$P(B|A) = P(B).$$

$$P(A \cap B) = 0$$

$$P(A \cap B)$$

$$= P(A)P(B)$$

Note: Events A and B **cannot be both independent and mutually exclusive** at the same time.

☑ **Example** 70% of your friends like chocolate, and 35% like chocolate and strawberry. What percent of those who like chocolate also like strawberry?

Let C and S denote events of liking chocolate & strawberry.

$$Q: P(S|C) = ?$$

$$P(S|C) = \frac{P(S \cap C)}{P(C)}$$

$$\text{Given: } P(C) = \frac{70}{100} = 0.7, \quad P(C \cap S) = \frac{35}{100} = 0.35$$

$$\therefore P(S|C) = \frac{P(S \cap C)}{P(C)} = \frac{0.35}{0.7} = 0.5$$

Example In a survey of 1000 people, 120 people have the flu. Out of those, 90 also have a sore throat. Among 880 people who do not have the flu, 110 have a sore throat. If a person is selected at random from this group, what is the probability that the person has the flu **given that the person has a sore throat**?

let F and S denote the events of having flu and sore throat

$$P(F|S) = ?$$

$$P(F|S) = \frac{P(F \cap S)}{P(S)}$$

$$\text{Given : } P(F \cap S) = \frac{90}{1000}, \quad P(S) = \frac{90 + 110}{1000}$$

$$\begin{aligned} \therefore P(F|S) &= \frac{P(F \cap S)}{P(S)} = \frac{\cancel{90}/\cancel{1000}}{\cancel{200}/\cancel{1000}} = \frac{200}{1000} \\ &= \frac{\cancel{90}}{\cancel{200}} = \frac{9}{20} \end{aligned}$$

Example A bag contains 3 red and 7 black balls. Two balls are drawn at random **without replacement**. If the second ball is red, what is the probability that the first ball is also red?

Let R and B denote events of red and black balls respectively.

$R_1 = 1^{\text{st}}$ ball to be red.

$R_2 = 2^{\text{nd}}$ ball to be red.

Q: $P(R_1 | R_2) = ?$

$$P(R_1 \cap R_2) = \frac{3C_1}{10C_1} \times \frac{2C_1}{9C_1}$$

$$= \frac{3}{10} \times \frac{2}{9} = \frac{1}{15}$$

$$P(R_1 | R_2) = \frac{P(R_1 \cap R_2)}{P(R_2)}$$

$$\# S_1 = 10C_1$$

$$\# S_2 = 9C_1$$

$$P(R_2) = P(\{1^{\text{st}} \text{ ball red and second ball red}\} \cup \{1^{\text{st}} \text{ ball black and second ball red}\})$$

$$= P(\{1^{\text{st}} \text{ ball red and second ball red}\}) + P(\{1^{\text{st}} \text{ ball black and second ball red}\})$$

$$= \left(\frac{3}{10} \times \frac{2}{9} \right) + \left(\frac{7}{10} \times \frac{3}{9} \right)$$

$$= \frac{1}{15} + \frac{7}{10} \times \frac{3}{9} = \frac{1}{15} + \frac{7}{30} = \frac{2+7}{30} = \frac{9}{30} = \frac{3}{10}$$

$$\therefore P(R_1 | R_2) = \frac{P(R_1 \cap R_2)}{P(R_2)} = \frac{1/15}{3/10} = \frac{1}{15} \times \frac{10}{3} = \frac{2}{9}$$

 **Example** Suppose that Jacob has O^+ blood with genotype OO/Rr , and Anna has B^- blood with genotype BO/rr . What is the probability that a child of Jacob and Anna will have B^+ blood?

